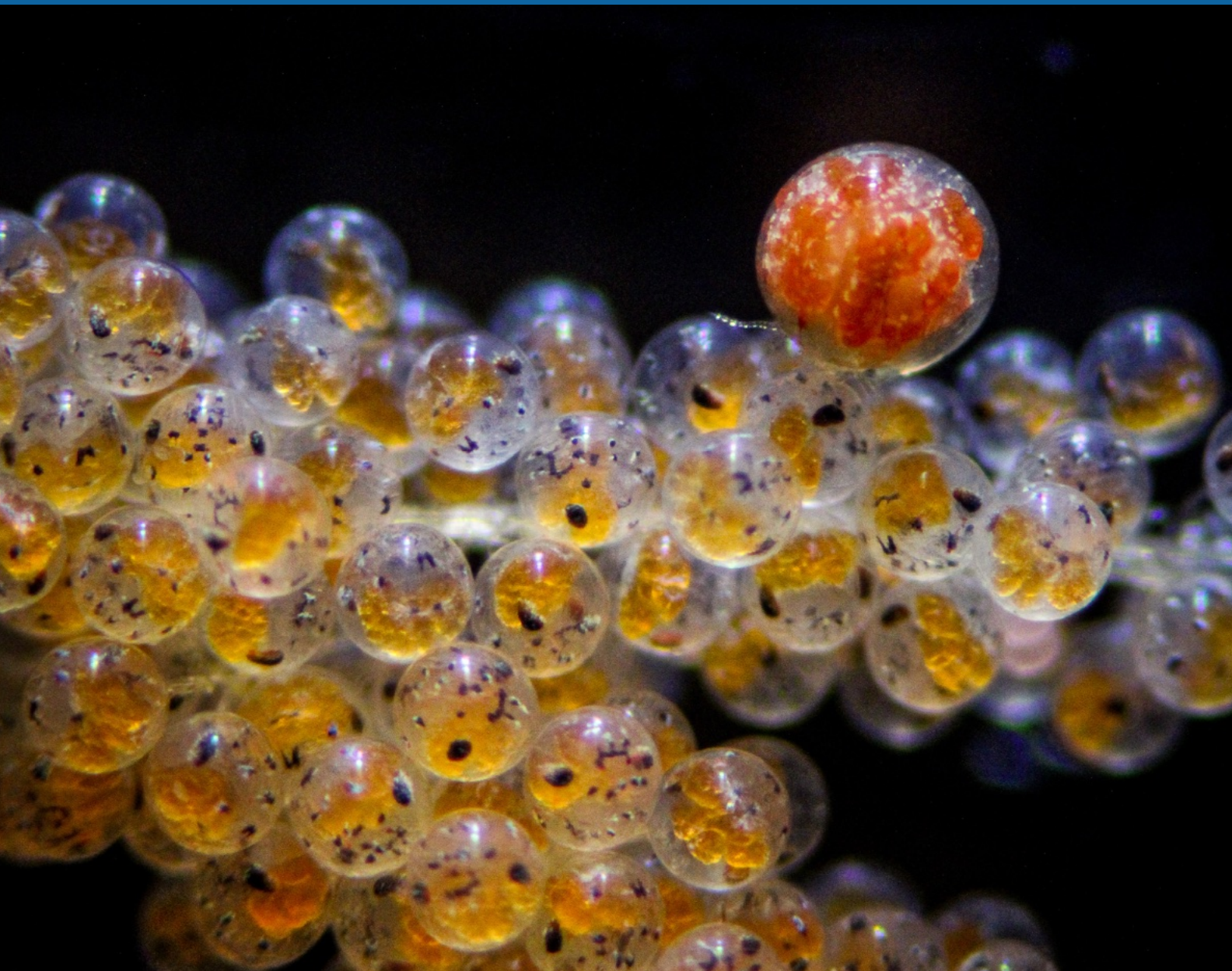


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COVER PHOTO: An adult copepod (*Choniosphaera* sp.) within the host (*Metacarcinus anthonyi*) egg mass. Female copepods appear highly modified, mimicking the coloration and globular shape of crab eggs. This previously undescribed egg predator was discovered in October 2021 during a laboratory course at the University of California, Santa Barbara, and is documented by Orli et al. (*Ecology*, Volume 106, Issue 12, Article e70263; doi:10.1002/ecy.70263). Nicothoid egg predators have never been observed in crabs along the Pacific coast of the Americas, despite extensive historical work on cancrid crab brood mortality in the 1990s. Photo credit: Jaden E. Orli.

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Discovery of an unidentified species of nicothoid copepod infesting cancrid crabs in Santa Barbara, California

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Externally brooding crustaceans host a variety of symbiotic egg predators, often causing substantial brood mortality (Kuris et al., 1991; Kuris & Wickham, 1987). Taxa known to live and feed on crustacean eggs (summarized by Kuris, 1993) include various microorganisms, rotifers, gastropods, worms (nemertean, turbellarians, nematodes, polychaetes, and oligochaetes), and other small crustaceans (copepods, amphipods, and isopods). Heavy infestations of egg predators, for example, nemertean worms in the Dungeness crab, *Metacarcinus magister*, have been implicated in the collapse (Wickham, 1979, 1986) and slow recovery (Hobbs & Botsford, 1989; Kuris et al., 1991) of some crustacean fisheries. The copepod family Nicothoidae includes parasites and symbiotic egg predators of other crustaceans, most of which have adopted egg mimicry as a life history strategy (Boxshall & Halsey, 2004). High intensities of nicothoids have negatively impacted the fecundity of commercially important species, such as the blue sand crab in Australia (Shields & Wood, 1993).

In October 2021, during a routine classroom laboratory activity at the University of California Santa Barbara focused on the demonstration of crustacean egg predators, we observed, for the first time, a nicothoid copepod in the genus *Choniosphaera* Connolly, 1929 infesting ovigerous yellow rock crabs (*Metacarcinus anthonyi* Rathbun, 1897). This discovery prompted further investigation, during which crabs were collected by fishermen from a 66–100 m depth using baited crab pots along the Gaviota Coast, west of Santa Barbara, CA (see Appendix S1: Section S1 for full methods). Crabs were held in flow-through aquaria during the investigation. We discovered *Choniosphaera* sp. in abundance on two other commercially important rock crab species, *Romaleon antennarium* Stimpson, 1856 and *Cancer productus* Randall, 1840. These crabs are habitat generalists, occurring in coastal waters from 0 to 150 m, and together form a small fishery in California (CDFW, 2019; Morris et al., 1980). Four species of nicothoids have been reported in association with seven species of brachyuran crabs, with varying effects

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on host fecundity (Bloch & Gallien, 1933; Connolly, 1929; Dang et al., 2022; Fischer, 1956; Gnanamuthu, 1954; Johnson, 1957; Santos & Björnberg, 2004; Shields & Wood, 1993). Nicothoid copepods have not been previously reported infesting decapod species from the Eastern Pacific (Appendix S1: Figure S1) despite extensive inspection of cancrid egg clutches during studies of the egg-predatory nemertean *Carcinonemertes epialti* Coe, 1902 in the 1990s (Shields et al., 1990, 1991). Furthermore, the University of California Santa Barbara Invertebrate Biology undergraduate class and instructors have carefully inspected cancrid egg masses for symbiotic egg predators annually since 1978. This study documents the first detection of a nicothoid copepod in these examinations. Members of two nicothoid genera, *Choniosphaera* and *Carcinothoe*, are reported to infest brachyuran crab egg masses (Bloch & Gallien, 1933; Connolly, 1929; Fischer, 1956; Lee & Kim, 2024). The nicothoid discussed herein is more closely aligned with *Choniosphaera* based on adult female morphological characteristics, including a globular shape with no abdomen, distinctly ventrally positioned mouthparts, a biarticulate exopod and uniarticulate endopod on legs 1 and 2, antennules with 11 segments, and a caudal ramus with four to five setae (Bloch & Gallien, 1933; Connolly, 1929; Lee & Kim, 2024).

The adult female nicothoid was the first life stage we observed on crab eggs (Figure 1a). We did not observe any adult nicothoids that we could recognize as male, and the apparent lack of male *Choniosphaera* sp. warrants further investigation. Adult females of *Choniosphaera* sp. appear to live permanently on the egg masses of their crab hosts and as such, their morphology is highly modified. The adult female body is globular and not visibly segmented. Its shape is similar to the host's eggs, and it camouflages well with the egg mass (Figure 1b; Appendix S1: Figure S3). Adult female *Choniosphaera* sp. change coloration as they increase in size and become gravid (Figure 1, see also Adult_Nicothoid_Video.MOV in Orli et al., 2025b). Adult nicothoids found on host egg masses that are in early developmental stages range in color from yellow to orange, while adults found on later stage egg masses are maroon (Figure 1a, see Appendix S1: Table S1 for a description of host egg developmental stages). We observed that adult female *Choniosphaera* sp. increase in size without molting, which is a phenomenon that has been documented for other symbiotic copepod females (Kabata, 1979; Ohtsuka et al., 2004, 2005; Smith & Whitfield, 1988). When adult females of *Choniosphaera* sp. were removed from the host eggs, all died within 24 h despite being kept well aerated in flow-through seawater and provided with eggs to feed on.

The *Choniosphaera* sp. life cycle appears to be coupled with host crab egg development (Appendix S1: Figure S2). Throughout the duration of host egg development (approximately 45 days for *M. anthonyi*; see Appendix S1: Table S1), adult females of *Choniosphaera* sp. oviposit egg sacs, carrying between one and four egg sacs at a time. As the females move throughout the host egg mass, their egg sacs become tangled among the host's eggs (Figure 1d). Each sac contains one to four embryos (Figures 1b and 2a). We were unable to track the number of eggs deposited per female throughout host egg development because it was impossible to distinguish individual nicothoids in situ, and the tendency of females to die when removed from the host's egg mass prohibited prolonged observation ex situ. The nicothoid copepod examined in this study follows a life cycle similar to the other two *Choniosphaera* species, *Choniosphaera cancrorum* and *Chonios. maendis* (Bloch & Gallien, 1933; Connolly, 1929). Of the 22 genera in the family Nicothoidae, only two genera, *Choniosphaera* and *Choniomyzon*, contain species where eggs hatch into nauplii (Wakabayashi et al., 2013). Once nauplii hatch from the egg sac (Figure 2b), they develop more appendages and increased segmentation through an unknown number of sequential molts until they reach the first copepodid stage, which is characterized by well-developed swimming legs (Figure 2d). The density of copepodids peaks near the end of host egg development (Appendix S1: Figure S2). Copepodids were observed feeding on host eggs.

Once the crab zoeae hatch from the host crab egg mass, the female crab removes old egg shells, aborted eggs, and debris from her pleopods (as noted by Shields et al., 1991), presumably including all nicothoids remaining on the egg mass. However, we also observed copepodids among host crab gill lamellae both before and after host egg hatching (Figure 2e). Other symbiotic egg predators of crabs (Kuris, 1993) have been reported from host gills, notably including a nicothoid, *Carcinothoe indica*, that is closely related to the species described herein (reported as *Choniosphaera indica* in Shields & Wood, 1993). We detected nicothoids in the egg masses of seven crabs that oviposited a subsequent clutch of eggs while under observation. Copepodid larvae were observed in these crabs' egg masses within a day of oviposition, and adult *Choniosphaera* sp. were present within 9 days (Appendix S1: Figure S2). Based on these observations, we speculate that copepodids migrate to the gills to wait for their host's next oviposition, but we did not observe copepodids moving from the host egg mass to the host gills. In at least one other nicothoid species, subadult females have been observed migrating through the host

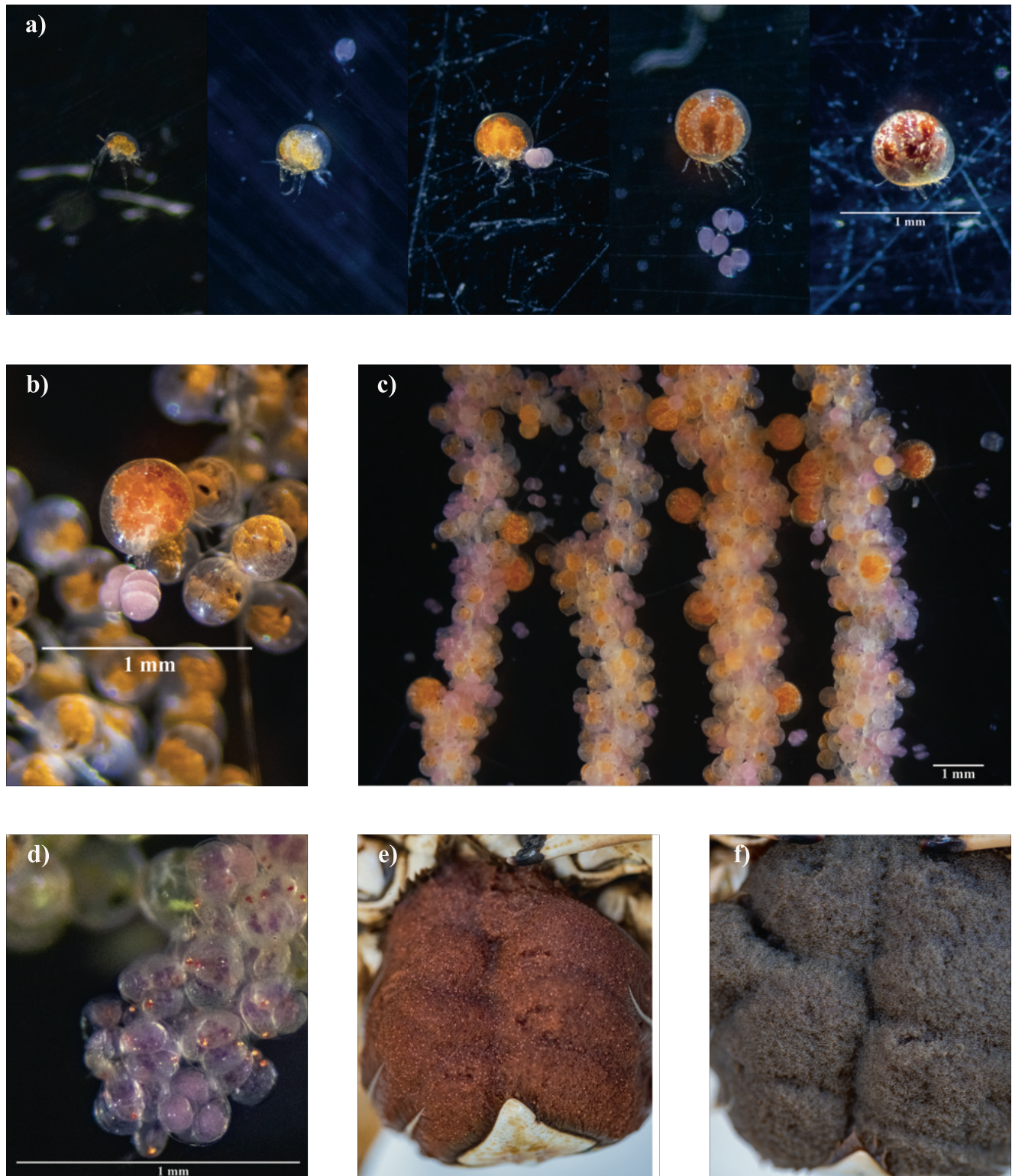


FIGURE 1 Adult nicothoids from *Metacarcinus anthonyi*; (a) sequence of sizes and colors of adult nicothoids; (b) adult nicothoids and their eggs (pink) among crab egg fascicles and several empty crab eggs whose embryos were consumed by nicothoids are also visible (viable crab eggs are distinguished by orange color and eyespots); (c) gravid adult nicothoid with egg sacs; (d) nicothoid egg sacs containing up to four embryos in each sac, note their red eyespots; (e) heavily infested crab egg mass, appearing maroon due to the great number of pink nicothoid eggs; (f) crab egg mass at the same developmental stage as (e), but with a comparatively low infestation of nicothoids. Photo credits: J. E. Orli.

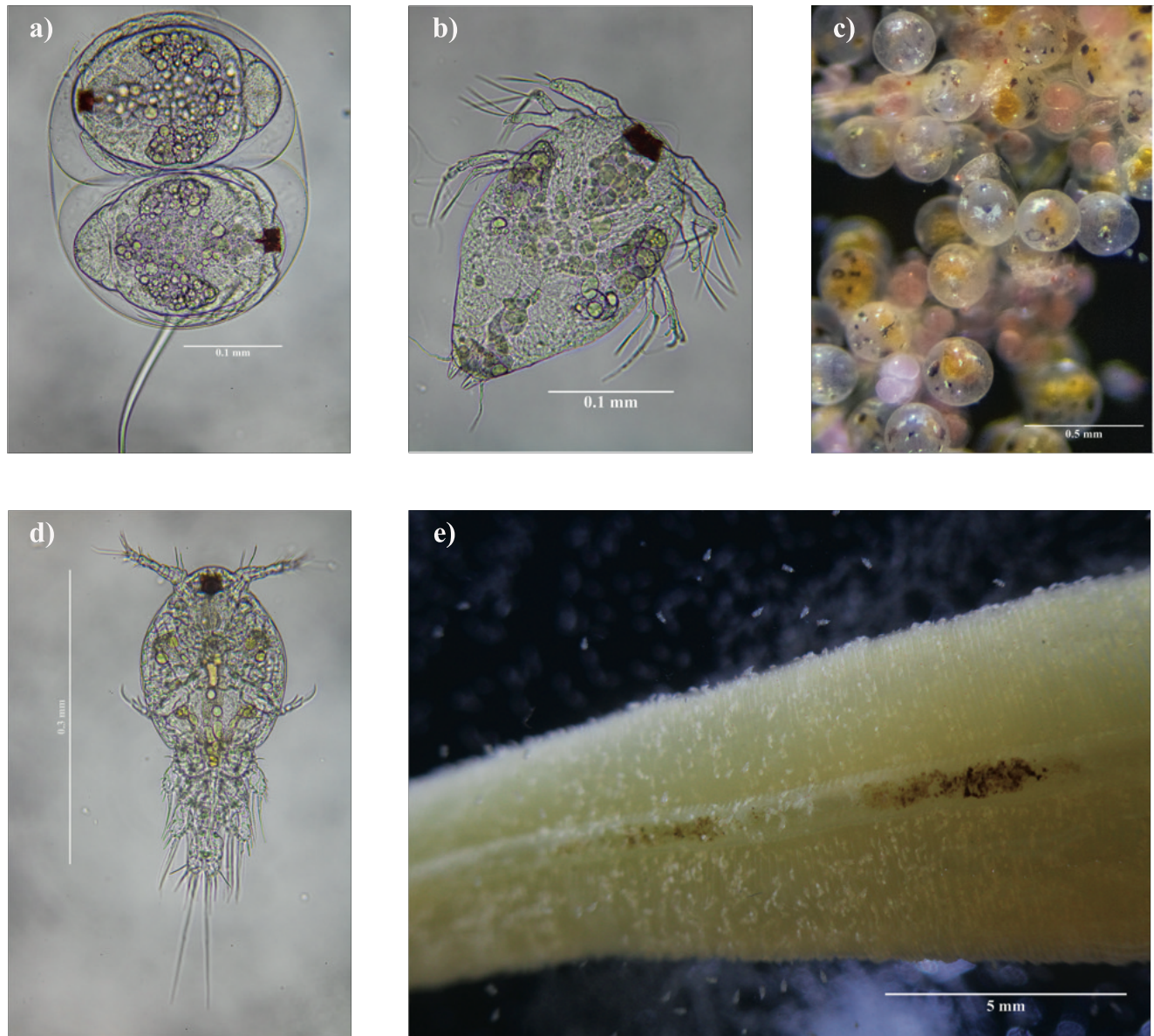


FIGURE 2 Larval nicothoid copepods taken from *Metacarcinus anthonyi* egg masses and gills; (a) two embryos within an egg sac, about to hatch into nauplii within an egg sac; (b) nauplius larva; (c) various nicothoid copepodids in situ in host egg mass. Note the red eye spots; (d) copepodid larva; (e) gill excised from *M. anthonyi* infested with nicothoid copepodids. Photo credit: J. E. Orli. (a–d) and G. O. Plewe (e).

body (Ohtsuka et al., 2007). We also recovered copepodids in the gills of male host crabs, suggesting that the copepodid stage may be able to disperse to new hosts and that male crabs may serve as reservoir hosts (Appendix S1: Table S2). The nauplii could also potentially disperse, but because of their lack of developed swimming legs, they would likely require physical contact between hosts to do so (Figure 2b).

We observed *Choniosphaera* sp. in *R. antennarium*, *M. anthonyi*, and *Can. productus* egg masses from Santa Barbara, CA, and San Diego, CA. Nearly all crabs of both sexes, for all three species examined, had at least one life

stage of nicothoid in their gills, their egg masses, or both (Appendix S1: Table S2). All ovigerous *M. anthonyi* and the majority of the other two ovigerous cancrids examined had reproductive nicothoid adults present in the egg mass. The only crabs uninfested by any nicothoid life stages (including the gills) were two juveniles, a non-ovigerous female, and two males that had been housed for over 6 months (Appendix S1: Table S2). It is possible that copepodids cannot survive in the gills indefinitely and that they must feed on egg yolk in order to reach adulthood, which would explain the observed loss of infection in male crabs. The lack of nicothoids in

juvenile crabs leads us to hypothesize that nicothoids do not colonize potential hosts until the crabs are sexually mature.

It is noteworthy that a distinctive nicothoid egg predator suddenly appeared in abundance on cancrid crabs in the Santa Barbara area. Nicothoid copepods had never been observed despite their distinctive appearance. This discovery suggests two mutually exclusive explanations: either this is an exotic introduction, or it is a native species which had previously existed unobserved on other crabs and has now transferred to a new group of hosts. The exotic hypothesis is unlikely because no congeneric nicothoid is known from the Indo-Pacific region, temperate, or boreal Pacific waters. A species in the similar genus *Choniomyzon* (D. Tang, personal communication) is common on *Portunus pelagicus* from India to Australia. However, historical studies on the two Atlantic species of *Choniosphaera* best support the hypothesis that nicothoids are native but perhaps ephemeral, emerging in abundance episodically over a time scale of decades. *Choniosphaera maenadis* was discovered on *Carcinus maenas*, the common and well-studied European shore crab, at Wimereux, Normandy in the mid-1930s (Bloch & Gallien, 1933; Gallien & Bloch, 1936), and reported from Whitstable, England, about 100 km from Wimereux by Gordon (Needham, 1933). It was not reported again until the 1950s from the North Sea, Germany (Fischer, 1956), and Norfolk, England in 1959 (Hamond, 1973). In both instances, *Chonios. maenadis* appears to have become locally abundant only to never be reported again (see also Stentiford, 2008). Similarly, in New England and the Canadian Maritimes, *Chonios. cancrorum* has been reported only twice, many years apart (Connolly, 1929; Johnson, 1957).

The presence of *Choniosphaera* sp. represents a threat to the under-managed and data-poor California rock crab fishery (Fitzgerald et al., 2018). An egg predator that is present in 100% of the breeding host population will likely impact population fecundity (see Appendix S1: Section S2 for further discussion). At the time that *Chonios. cancrorum* was discovered in the brood of *Carcinus maenas* in New England, the effect of *Chonios. cancrorum* was notable enough to garner hope for its use as a biological control of that invasive pest species (Johnson, 1957). The discovery of this novel egg predator in Southern California could have a deleterious effect on the Santa Barbara rock crab fishery and the general role of these cancrids as predators in nearshore subtidal habitats. Notably, these nicothoids add to brood losses caused by *Carcinonemertes* sp. nemertean egg predators (Shields et al., 1990). The sudden emergence of an epizootic of

nicothoid egg predators combined with a robust population of existing egg predators warrants further research on their impact on the fecundity of host populations. Our research team is currently quantifying egg mortality rates in infested egg masses. Further research should focus on revealing the current geographical extent of the nicothoid outbreak and the diversity of its host range among Eastern Pacific crabs.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data and code (Orli et al., 2025a) are available in Dryad at <https://doi.org/10.5061/dryad.rv15dv4fw>. Video files (Orli et al., 2025b) are available in Zenodo at <https://doi.org/10.5281/zenodo.10699477>.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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Appendix S1

Journal: Ecology

Manuscript Title: Discovery of an unidentified species of nicothoid copepod infesting cancrid crabs in Santa Barbara, California

Authors: Jaden E. Orli, Sophia M. Lecuona, Gabrielle O. Plewe, Carson N. Gadler, Armand M. Kuris, Danny Tang, Zoe L. Zilz

Section S1. METHODS

S1.1. Crab Collection and Housing

After the initial observation of nicothoid egg predators in *Metacarcinus anthonyi* egg masses in 2021, we obtained rock crabs from baited crab pots set by local fishermen along the Gaviota Coast of Santa Barbara County to determine the prevalence and abundance of the nicothoid. We collected data from 65 ovigerous females (54 *M. anthonyi*, 4 *Romaleon antennarium*, and 7 *Cancer productus*), 23 males (12 *M. anthonyi*, 7 *R. antennarium*, and 4 *C. productus*), and 1 juvenile female *C. productus*. Crabs were housed in individual tanks with an independent source of running filtered seawater. Upon collection we recorded carapace width to 1 mm, abdomen width to 1 mm, weight the nearest gram, injuries, and initial egg development stage or EDS (Appendix S1: Table S1).

S1.2. Host Crab Brood Examination

All ovigerous female crab egg masses ($n = 65$) were routinely examined for the presence of nicothoid copepods. Every three days we collected a sample of at least one thousand crab eggs from the center of a haphazardly selected pleopod by removing a few fascicles of setae with attached eggs. The total number of crab eggs per sample was counted and empty eggshells and eggs with evidence of arrested development were quantified. This process was repeated for each ovigerous female until the crab hatched her brood. Most post-ovigerous crabs were released but

some were dissected to inspect the gills and other internal organs for nicothoids (see Host Crab Dissections). Halfway through egg development (EDS E and later; Appendix S1: Table S1) we also quantified nicothoid egg sacs and larval/juvenile stages. We calculated the density of nicothoids per crab egg and then averaged per EDS. Because *Metacarcinus anthonyi* fecundity ranges from 700,000 to 3,300,000 eggs per brood (Shields et al., 1990) we extrapolated density to calculate abundance of nicothoids per 1,000,000 eggs to estimate total nicothoid abundance per crab (Appendix S1: Figure S2). To observe movement and feeding behavior of the nicothoids, we placed a single adult nicothoid with a fascicle of 100 to 200 host eggs in a mesh container with flowing seawater until that nicothoid died, which usually occurred within a few hours.

S1.3. Host Crab Dissections

To determine the presence of the nicothoid in other Southern California crab species, we dissected 28 *M. anthonyi*, 8 *R. antennarium*, and 7 *C. productus*. Crabs were humanely euthanized by pithing through the ventral thorax directly over the thoracic ganglion. We removed the two center gills from both the left and right side of the crab and placed them in petri dishes with seawater. Each gill was examined under a dissecting microscope for nicothoids, damage or scarring, and the accumulation of a microbial biofilm. We randomly selected one gill, counted the lamellae, and then rinsed the gills with ethanol to immobilize any attached nicothoids which allowed us to count them. We recorded the number and life stage of nicothoids per gill lamellae. Samples of other organs, including gonads and digestive tissue, were also examined for the presence of nicothoids.

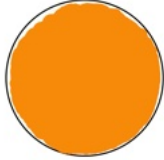
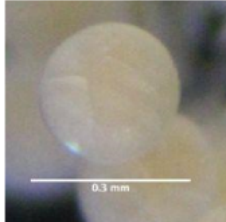
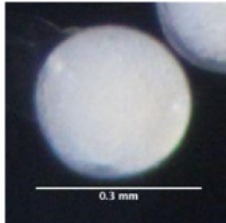
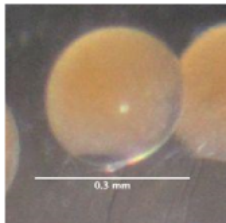
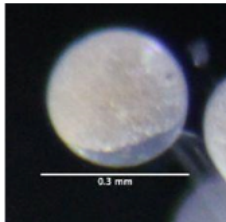
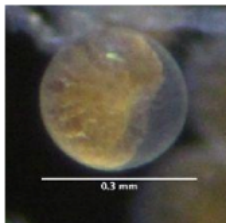
Section S2: POTENTIAL IMPACTS ON HOST REPRODUCTION

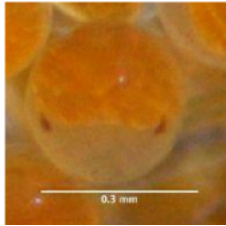
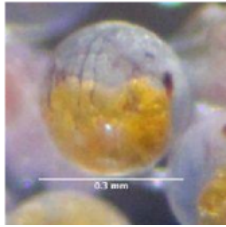
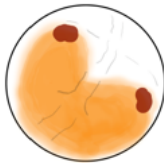
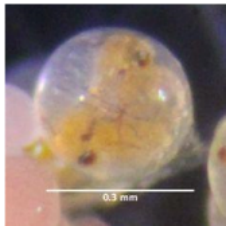
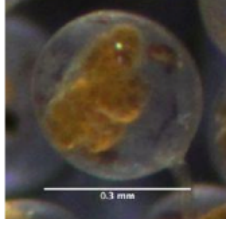
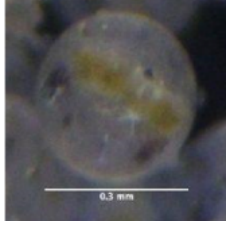
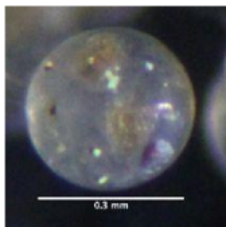
Based on our observations of nicothoid feeding activity

([Adult Nicothoid Feeding Video](#).mov in Orli et al. 2025) and its impact on host eggs (Main Text: Figure 1b) we hypothesize that nicothoids can have a substantial effect on the fecundity of individual crabs. Although feeding rates of adults or copepodids have not yet been quantified, we document high infestations of nicothoid adults resulting in near total loss of host eggs on some ovigerous hosts. Nicothoids become so numerous that they can effectively replace host eggs with their own eggs ([Infested Host Eggs Video](#).mov in Orli et al. 2025). Copepodid presence in the gills of non-gravid crabs and the prompt appearance of copepodids in newly oviposited egg masses suggests that copepodids migrate to the gills in between broods, returning to the host eggs upon subsequent oviposition. This behavior allows for a female crab to be infested for the entirety of her reproductive period. The larval nicothoids present in the gill chamber presumably function as the baseline population for infestations in future, subsequent, broods (Appendix S1: Table S1). Samples from subsequent broods showed higher abundances of adult nicothoids and nicothoid eggs than on prior broods. We examined crabs immediately post-molt and recovered copepodids from the gills, indicating that the crabs retain infestations following a molt, but the mechanism of this retention is unknown. Uninfested female crabs may initially become infested upon contact with an infested male crab. Coupled with the continuous reproduction of *M. anthonyi* and *R. antennarium* throughout the year (Shields et al. 1990), continuous infestation from the gill sites likely contributes to the proliferation of this novel nicothoid epizootic ([Highly Infested Host Gills Video](#).mov in Orli et al. 2025). No negative

impact on the host crab has been observed as a result of the presence of the copepodids in the host gill chambers, but highly infested gills, like those we observed, potentially result in reduced gas exchange (Main Text: Figure 2e, Appendix S1: Table S1).

TABLE S1. *Metacarcinus anthonyi* crab egg development stages (EDS). Illustrations by Z. L. Zilz and photographs by J. E. Orli.

Stage	Description	Average Duration (days)	Sketch	Image
A	Between 1 and 64 cells, all yolk. 10% developed.	4.1 days		
B	One small clear spot visible.	3.7 days		
C	2 small clear spots visible.	1.7 days		
D	One large spot is visible, but no segmentation visible yet. About 20% developed.	2.6 days		
E	Segmented appendages (tagmata) visible below large clear spot. About 40% developed.	11.9 days		

F	Eyespots visible for the first time. 55% developed.	2.0 days		
G	Abdominal pigment visible. 60% developed.	2.3 days		
H	Yolk has started to divide into lobes, heartbeat is visible. 67% developed.	1.7 days		
I	Yolk is 4 lobed. 75% developed.	6.9 days		
J	Yolk is two-lobed but lobes are still connected. 85% developed.	2.0 days		
K	Yolk is two lobed and separated. 90% developed.	2.1 days		

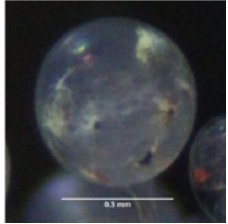
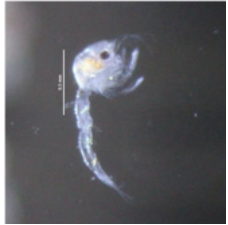
L	Yolk absent, lots of “flexure” i.e. movement within the egg casing.	1.5 days		
M	Active hatching.	0.8 days		

TABLE S2. Summary of prevalence and intensity of feeding nicothoid stages (copepodids and adults) in male and female crab hosts, by species.

Species	Sex	Prevalence Infested	Prevalence with Adults in Host Egg Mass	Mean Intensity of Copepodids in Host Gill Chamber
<i>Metacarcinus anthonyi</i>	F	100% (n = 55)	100% (n = 54)	14,282 ± 21,755 (n = 19)
	M	83% (n = 12)	NA	235 ± 373 (n = 10)
<i>Cancer productus</i>	F	66% (n = 9)	100% (n = 4)	24 ± 31 (n = 3)
	M	100% (n = 4)	NA	42 ± 59 (n = 4)
<i>Romaleon antennarium</i>	F	66% (n = 6)	100% (n = 2)	4,608 ± 5,578 (n = 4)
	M	100% (n = 8)	NA	61 ± 34 (n = 7)

Notes: Figures are means ± standard deviation. Individuals were considered infested if any nicothoid life stage (adult, copepodid, nauplius, or egg sac) was confirmed to be present during any evaluation. The intensity of copepodids in the host gill chamber was scaled to a total of 12 host gills.

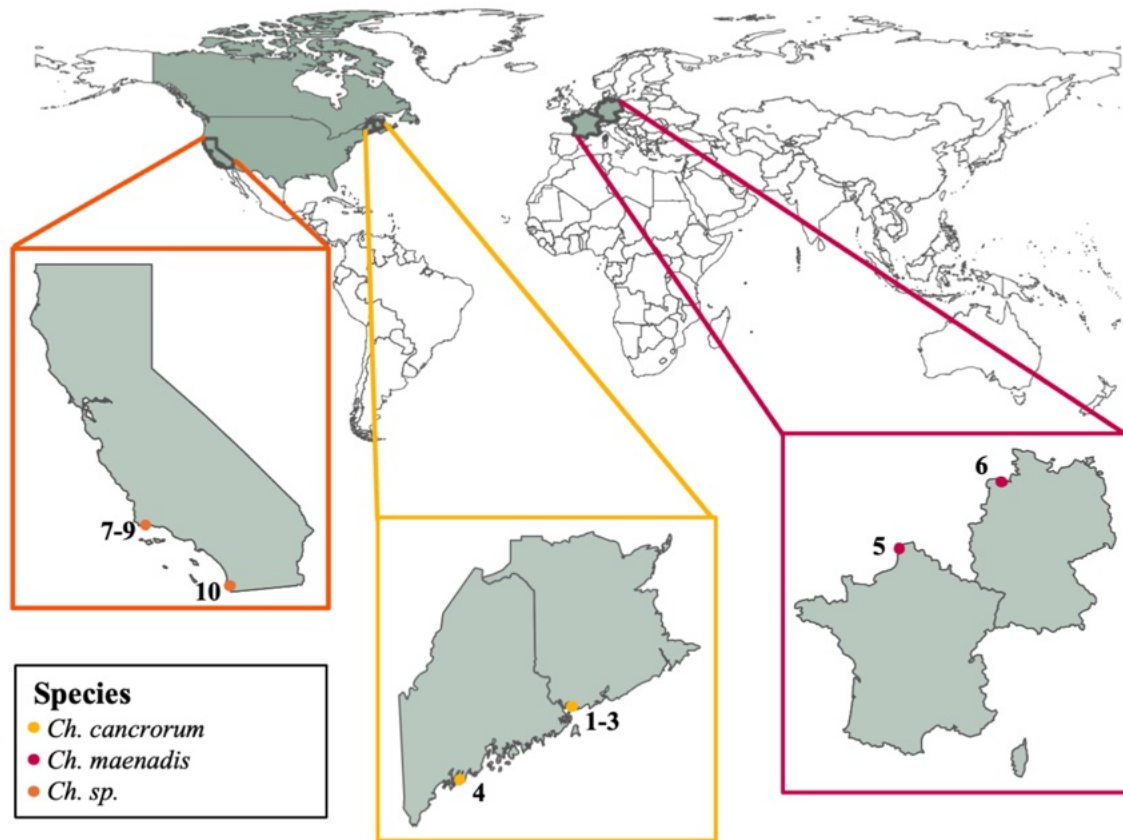


FIGURE S1. Global distribution of species in the genus *Choniosphaera*, the most likely parent genus of the nicothoid reported in this publication. Individuals of this genus have previously only been identified in the Atlantic Ocean. Numbers indicated on the map correspond to the first documentation of each *Choniosphaera* species infesting a novel host species at that location. Individuals of this genus were found infesting (1) *Cancer irroratus*, (2) *Ateacyclus undecimdentatus*, and (3) *Cancer borealis* in Passamaquoddy Bay (Connolly, 1929), (4) *Carcinus maenas* in Linekin Bay (Johnson, 1957), (5) *C. maenas* in the English Channel (Bloch and Gallien, 1933), (6) *C. maenas* in the Wadden Sea (Fischer, 1956), in the Pacific Ocean; (7) *Metacarcinus anthonyi*, (8) *Cancer productus*, and (9) *Romaleon antennarium* on the Gaviota Coast (this publication), and (10) *M. anthonyi* in San Diego (this publication).

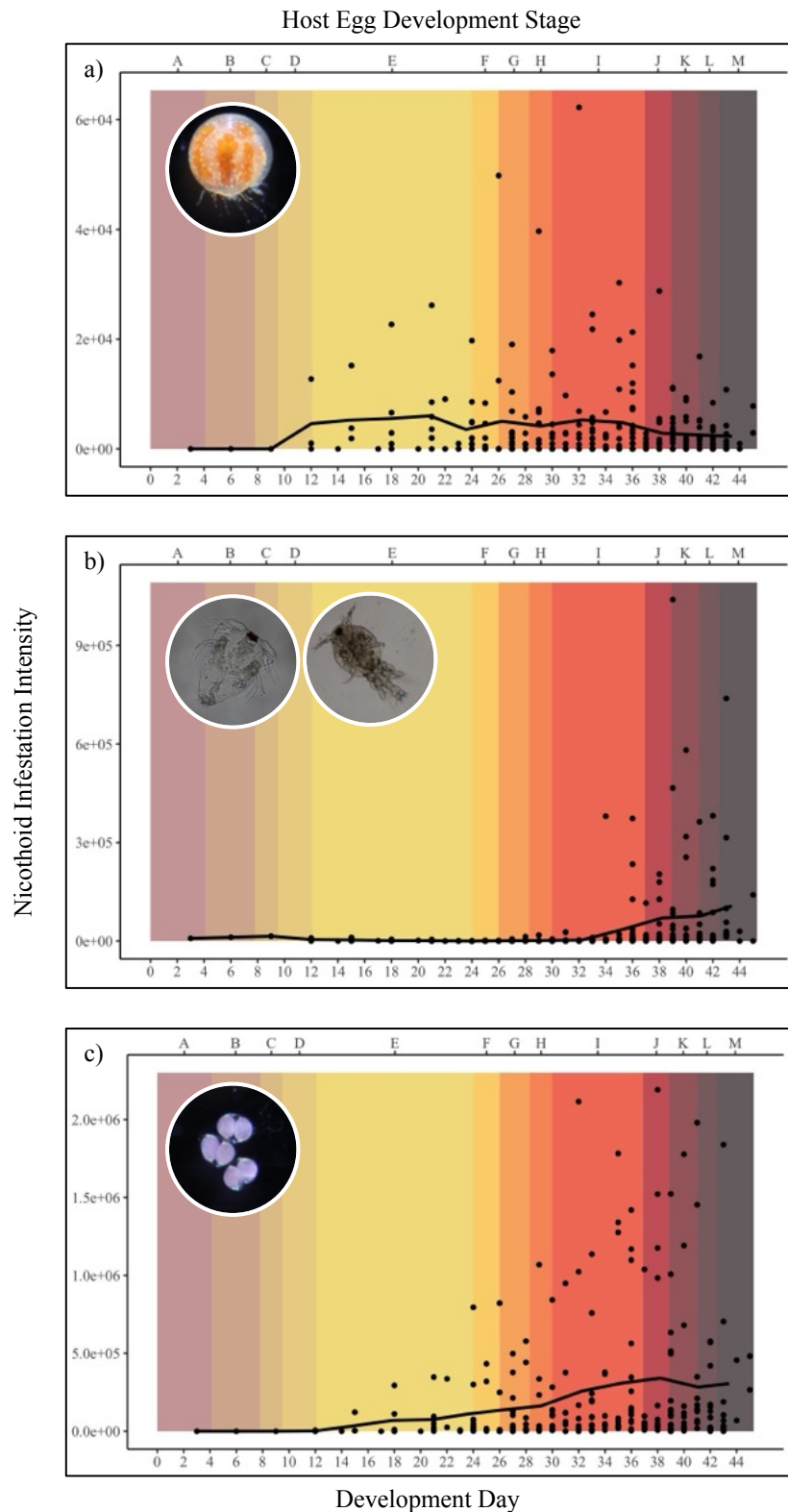


FIGURE S2. Estimated nicothoid intensity throughout the duration of host crab brooding period for (a) adult nicothoids (n = 41 crabs), (b) larval nicothoids (n = 41 crabs), (c) nicothoid eggs (n = 41 crabs). Points represent estimated nicothoid intensities for an individual crab on a particular “development day” (day since host-egg deposition). The solid lines represent the mean estimated intensity of nicothoids per crab egg clutch. Nicothoid intensity was estimated by calculating the density of nicothoids per host egg, averaging every three days, and then scaling to one million host eggs to approximate intensity in an entire host brood. Colored bars correspond to the host egg development stages outlined in Table S1. Some individual crabs were only examined for a portion of their brooding duration. Photo credits J. E. Orli.

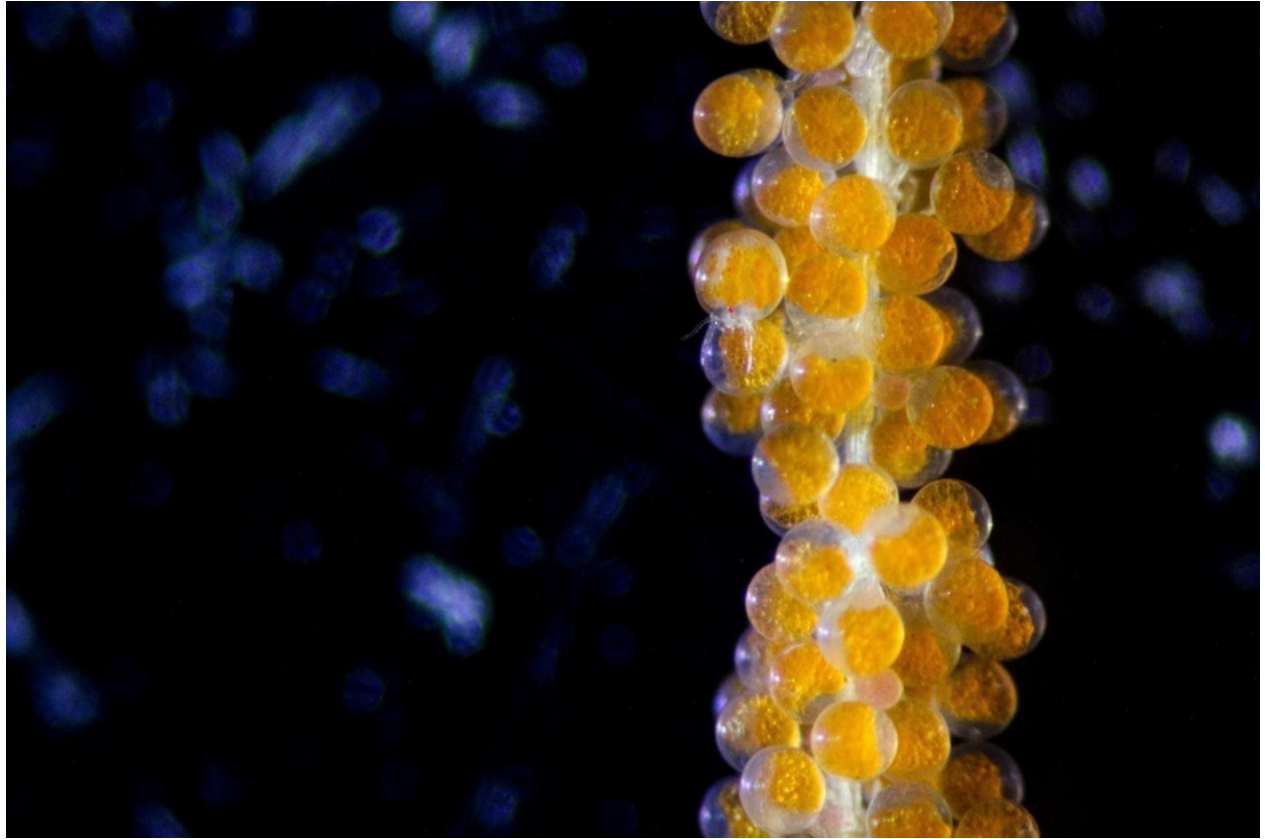


FIG S3. Adult nicothoid camouflaged amongst host crab eggs. Photo credits: J. E. Orli.

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